

Steamboat Hills: Talking Points for Groundwater Hydrology Advisor Meeting

Where the data stands, and how to think about modeling the system

2026-10-01

Why this conversation, right now

Over the last few weeks a set of scanned NDEP field-sampling forms got transcribed into the project database, turning a handful of one-off grab samples into a real, roughly **monthly time series (late September through December 2025) across 14 sites** – six named injection-side ports (Galena 1/2/3, SB2, SB3, SBHR) and eight shallow monitoring/domestic wells (Soccer Field, Herz Domestic, Herz Deep, Eich, NDOT, Boyd, Jeppson, Rogers). This is the densest *repeated* field-parameter dataset the project has had to date, and it raises a concrete question worth a faculty opinion: **what modeling approach does data like this actually support?**

What the data shows

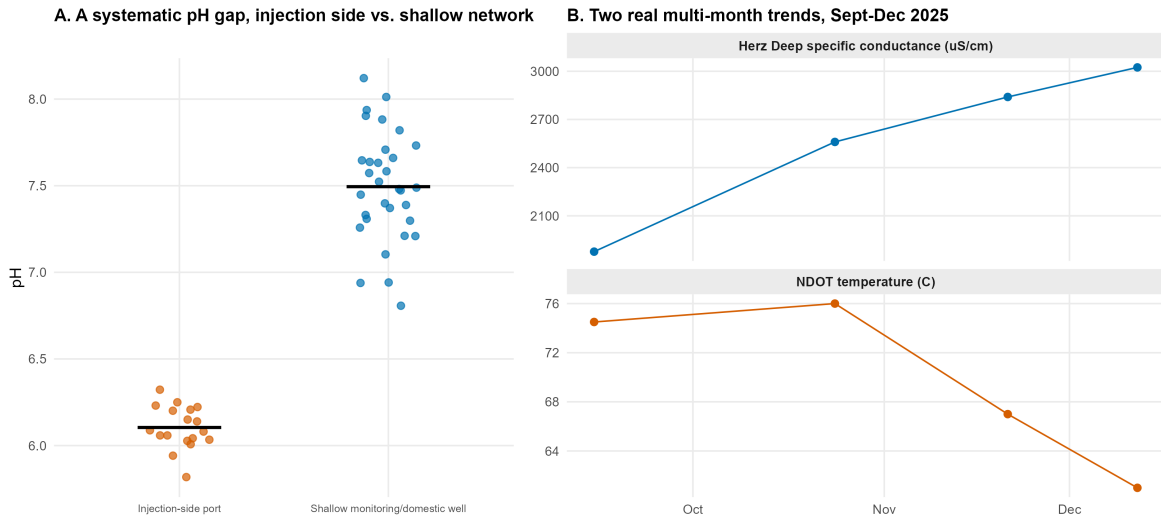


Figure 1: Left: pH across the six injection-side ports vs. the eight shallow monitoring/domestic wells, pooled across three real monthly rounds. Right: two real multi-month trends at individual wells, Sept-Dec 2025.

1. A stable pH gap between the injection side and the shallow network. Every one of the six injection-side ports reads pH 5.8-6.3 at every round; every one of the eight shallow wells reads pH 7.0-8.1. That ~1.5-2 unit gap holds across all three rounds, not just one snapshot – a real, repeatable signature, not noise.

2. Two individual wells show real, opposite-direction trends over the same three months. Herz Deep's specific conductance rises from 1,877 to 3,024 uS/cm (roughly +60%) while its temperature stays essentially flat (~30-36 C). NDOT's temperature falls from about 76 C to 61 C while its conductance stays roughly flat (~3,000-3,500 uS/cm). Neither well's two parameters move together.

3. A broader context for these 14 sites: this project's own poster finding puts current thermal-water discharge to Steamboat Creek at roughly 27-59 L/s (a single real paired Cl-discharge measurement gives 58.7 L/s; the poster's own multi-date average is ~27 L/s), compared to a historical range of roughly 60-90 L/s reported for 2008-2016 and over 1,000 L/s reported by White (1968) in the 1950s-60s – the long-run context this new monthly dataset sits inside.

Questions for the conversation

- **Does the pH gap argue for reactive transport over conservative mixing?** A stable, multi-round pH offset between an injected fluid and the shallow network looks like water-rock buffering along a flow path – is that the right read, or could something simpler (e.g. a measurement-method difference between the two sample types) explain it just as well?
- **How should two wells moving in opposite, uncorrelated directions be interpreted?** Herz Deep’s conductance rise and NDOT’s temperature fall don’t look like the same mechanism. Is this two independent, local processes, or a sign that neither temperature nor conductance alone is a safe stand-in for “thermal influence” at this site?
- **What’s the right first modeling step given the real data density we have?** A full 3-D reactive-transport model is the long-term goal, but real continuous water-level coverage *inside* the Steamboat field itself is currently limited to 2-3 wells (everything else with continuous water level sits in the wider South Truckee Meadows basin). Is a simpler lumped-parameter/box model (two- or three-end-member mixing with a time-varying fraction) the appropriate first step, or is it worth waiting on better spatial coverage before committing to any specific model structure?
- **Does this dataset make the planned Cl-conductivity calibration campaign more or less urgent?** The project’s sampling-frequency statistical framework (Monte Carlo subsampling, rolling-origin cross-validation) is fully built but still synthetic/proxy-only – no real chloride sample yet overlaps the conductivity-logger deployment window. Given the pH/trend patterns above, would a professor prioritize that calibration field season over, say, pursuing the fault-digitization/potentiometric-surface work in parallel?

One flagged data-quality note, for transparency

One value in this new batch (Herz Domestic Well, 10/24/2025 specific conductance, 9,933 uS/cm) is roughly 7-9x every other reading on file for that well and is flagged in the database as a likely OCR transcription error pending a check against the original scanned form – not used in the trends described above.

One-paragraph summary, if time is short

“We just turned scattered grab samples into a real three-month, 14-site monthly time series. Two patterns stand out: a stable pH gap between the injection fluid and the shallow groundwater network (suggesting real water-rock interaction along the flow path), and two wells trending in opposite directions on temperature vs. conductivity over the same window (suggesting neither parameter alone is a safe universal proxy for thermal influence here). I’d like your take on whether this points toward a reactive-transport framing over conservative mixing, and what

the right first modeling step is given that our real water-level coverage inside the field is still thin.”